

Yulu - Business Analysis Report

 Yulu_Hypothesis_Testing.pdf

<https://drive.google.com/file/d/1mTazNUbcTtNgewrAJCntzWQFQC4sCNYh/view?usp=sharing>

1. Executive Summary

Yulu, India's leading micro-mobility service provider, analyzed factors affecting demand for shared electric cycles. Key findings indicate that **weather, temperature, and hourly commuting patterns** drive most of the demand, while weekday/weekend and seasonal variations have smaller effects. Registered users dominate usage. Optimizing operations based on these insights can improve efficiency and revenue.

2. Key Insights

1. Consistent Demand Across Weekdays and Weekends

- Daily rentals do not vary significantly between working days and weekends.
- Suggests commuting patterns are stable throughout the week.

2. Weather is a Major Driver of Demand

- Clear and partly cloudy days account for ~66% of rentals.
- Misty/cloudy conditions account for ~26%.
- Rain and snow days drastically reduce rentals (very few rentals (~8%)).

3. Seasonal Trends Exist but Are Mild

- Slightly higher rentals in winter, fall, and summer; lowest in spring.
- Seasonal variation is noticeable but not extreme.

4. Temperature and Comfort

- Rentals increase with moderate temperatures (~15–30°C) and drop when it gets too hot (>30°C).
- Perceived temperature aligns with comfort trends.

5. Humidity and Wind

- Very high humidity (>90%) reduces rentals.
- Strong winds are rare but temporarily decrease usage.

6. Hourly Peaks Reflect Commuting Patterns

- Morning peak: 8–9 AM
- Evening peak: 5–7 PM
- Early morning and late-night periods have low usage.

7. Registered Users Dominate the System

- Casual riders are few; focus should be on regular users for operations and marketing.

8. Balanced Data Across Months and Years

- Rentals are evenly distributed across months and years → reliable for trend analysis.

<u>Factor</u>	<u>Observation</u>	<u>Implication</u>
Weekday vs Weekend	Rentals similar across working days and weekends	Staffing and fleet allocation can remain stable throughout the week
Weather	Clear/Partly cloudy: ~66%, Mist/Cloudy: ~26%, Rain/Snow: negligible	Allocate more bikes on clear days; reduce load during adverse weather
Season	Slightly higher rentals in winter, fall, summer; lowest in spring	Minor seasonal adjustments to fleet and maintenance schedules
Temperature	Rentals increase up to ~30°C, then decline	Plan operations for comfort-driven demand

Humidity	High humidity reduces rentals	Monitor and schedule bike availability accordingly
Hourly Pattern	Peaks at 8–9 AM and 5–7 PM	Optimize bike placement and staff for peak commute hours
User Type	Registered users dominate	Focus on retention and loyalty programs for registered users

3. Recommendations

1. Operational Planning

- Maintain consistent bike availability and staffing across all days.
- Increase bike allocation on clear or partly cloudy days.
- Reduce operational load during adverse weather conditions.

2. Seasonal Strategy

- Minor fleet and staffing increases in winter, summer, and fall.
- Align maintenance and promotions with seasonal demand.

3. Promotions & Incentives

- Offer discounts during low-demand weather days.
- Target casual users to convert them into registered users.

4. Data-Driven Adjustments

- Monitor weather forecasts for fleet deployment decisions.
- Track hourly rentals to optimize peak-hour bike distribution.

5. Customer Comfort

- Consider heat and humidity in service planning.
- Offer promotions or alternatives during extreme weather.

4. Key Business KPIs

<u>KPI</u>	<u>Purpose</u>	<u>Target / Measurement</u>
Daily Rentals	Track total usage	Total rides per day (casual vs registered)
Weather-Adjusted Utilization	Optimize fleet based on weather	Rentals per weather type
Hourly Peak Utilization	Allocate bikes efficiently	Rentals per hour; monitor AM/PM peaks
Seasonal & Monthly Demand	Plan long-term operations	Average rentals per month/season
User Retention & Adoption	Grow registered user base	Ratio of registered to casual users
Operational Efficiency	Minimize downtime	Bike uptime %, staff utilization
Promotion Effectiveness	Maximize campaign ROI	Rentals generated from campaigns

5. Conclusion

- Weather and temperature strongly affect bike usage.
- Daily commuting patterns dominate; weekday/weekend differences are minor.
- Most rentals come from registered users.
- Applying these recommendations and tracking KPIs will **help Yulu optimize operations, improve customer satisfaction, and increase revenues.**

